

CSA1280 – COMPUTER ARCHITECTURE

EXPERIMENT – 23

LARGEST NUMBER IN AN ARRAY

AIM: To write and execute an 8085 assembly language program to find minimum number in an array and store result in memory.

ALGORITHM

1. Store the array elements in consecutive memory locations starting from **2050H**.
2. Load the number of elements into register **C**.
3. Assume the first element as the minimum number.
4. Compare the minimum number with the next element.
5. If the next element is smaller, update the minimum number.
6. Repeat until all elements are compared.
7. Store the minimum number at **2055H**.
8. Stop the program.

CODE:

```
MVI C,05H
```

```
LXI H,2050H
```

```
MOV A,M
```

```
INX H
```

```
DCR C
```

```
LOOP: CMP M
```

```
    JC NEXT
```

```
    MOV A,M
```

```
NEXT: INX H
```

```
    DCR C
```

```
    JNZ LOOP
```

STA 2055H

HLT

OUTPUT:

The screenshot displays the GNUSim8085 - 8085 Microprocessor Simulator interface. The main window is titled "GNUSim8085 - 8085 Microprocessor Simulator" and contains several panels:

- Registers:** Shows the state of various registers. The Accumulator (A) is 00, BC is 00 00, DE is 00 00, HL is 20 55, PSW is 00 00, PC is 42 16, SP is FF FF, and Int-Reg is 00. The Flag register (F) is 0.
- Decimal - Hex Conversion:** A section for converting between decimal and hexadecimal values.
- I/O Ports:** A section for managing I/O ports, currently showing port 0 with value 00.
- Memory:** A section for managing memory, currently showing address 2054 with value 06.
- Assembly Code:** A list of assembly instructions. The code is as follows:

```
1 MVI C,05H
2 LXI B,2050H
3 MOV A,M
4 INX H
5 DCR C
6
7 LOOP: CMP M
8       JNC NEXT
9       MOV A,M
10
11 NEXT: INX H
12       DCR C
13       JNZ LOOP
14
15 STA 2055H
16 HLT
```
- Memory Window:** A table showing memory addresses and data. The data is as follows:

Address (Hex)	Address	Data
0802	2050	12
0803	2051	20
0804	2052	25
0805	2053	8
0806	2054	6
0807	2055	6
0808	2056	0
0809	2057	0
080A	2058	0
080B	2059	0
080C	2060	0
080D	2061	0
080E	2062	0
080F	2063	0
- Assembler Message:** A message box showing the status of the assembly process. The message is "Program assembled successfully".